



ORIGINAL RESEARCH

Ensemble Machine Learning Models for Corporate Credit Risk Prediction: A Comparative Study

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Abstract

The objective of the study is to assess the predictive capabilities of four machine learning models such as XGBoost, CATBoost, LightGBM, and LSTM to identify the significant variables that determine whether a company is default or non-default in terms of credit risk. The study also applied two feature engineering techniques such as principal component analysis (PCA) and random forest to extract the important features and then applied the machine learning models. The analysis used a dataset from the Prowess database, which consists information on companies that are rated by the ACUTE and BRICKWORK credit rating agencies. The dataset comprised of 212 non-default-rated companies and 198 default-rated companies from 2016 to 2024. The study utilized 27 explanatory variables. The analysis showed that ensemble models, particularly the XGBoost, CATBoost, and LightGBM showed higher performance when the model was created using the features selected by the random forest algorithm compared to the models created with components created by PCA. However, LSTM showed higher performance with the components reduced using PCA when compared to the model created using feature selection with random forest. These models help both banking and non-banking financial institutions to make lending decisions, based on the borrower firm's creditworthiness. This research also helps in the selection of most important features which would help to identify additional insights to assess a firm's financial performance.

Keywords Credit risk · Defaulters · Machine learning · Random Forest · PCA · India

Introduction

Organizations in the Indian business sector must rely on strong credit risk evaluation systems owing to their dynamic and multifaceted operational characteristics. Indian businesses conduct their operations as a part of various sectors that include Information Technology and manufacturing alongside trade activities. A wide range of businesses including SMEs and corporations operate in the Indian environment which makes precise credit risk evaluations complex. The rising number of non-performing assets (NPAs) presents an additional challenge to the problem because most firms missed their debt payments during the recent period. Modern economic conditions along with increased non-performing assets (NPAs) have motivated financial institutions and non-financial enterprises to seek improved

credit risk management approaches for Indian financial risk reduction. The Indian government established various response approaches to solve business financial problems among firms facing economic struggles. Diverse law structures existed for institutions to handle debt-default cases during the pre-2016 era along with individual policies and regulations that served as foundation. From 1985 until 2002 the Sick Industrial Companies Act (SICA) maintained partnership with Debt Recovery Act of 1993 as primary debt management instrument in addition to the Corporate Debt Restructuring Mechanism and the Strategic Debt Restructuring Scheme. Standard guidelines formed part of the Insolvency and Bankruptcy Code when it was launched in 2016 to manage distressed assets. The National Company Law Tribunal delegates insolvent business supervision tasks to Insolvency Professionals through this framework. A business requiring bankruptcy must create either a rebuilding plan for current property conditions or proceed with complete asset disposal to resolve its business problems. The introduction of IBC by different firms allowed for insolvency proceedings and business liquidations. Studies about corporate

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financial failure drivers maintain high importance because their findings yield multiple practical applications. This study investigated the factors that influence corporate valuation as well as industrial financial stability of Indian firms. This research study validated the machine learning models through assessments of compliant data obtained from the company's databases. The research output will improve both risk insurance operations and financial institution loan application review activities. Indian financial institutions can use machine learning for credit risk analysis to build risk management systems that cut down losses and strengthen economic stability in the country.

Literature Review

This research analyzed published literature on machine learning algorithms that assist organizations forecast their credit risk levels. The study conducted by Emmanuel et al. [1] explained fundamental barriers in credit risk forecasting through their analysis of how traditional logistic regression techniques fail to detect non-linear patterns and deal with dataset distribution problems. The combination of Random Forest and Gradient Boosting as a first step followed by Extreme Gradient Boosting produced exceptional predictive results. The validation of classifiers maintains great importance in ensemble learning/-boosting due to the employment of Information Gain (IG) for selecting significant predictors and speeding up parallel algorithm execution. The proposed stacking method improved both performance and robustness of class identification by using base model prediction inputs within a meta-learner framework for imbalanced classes. For evaluating machine learning models that manage imbalanced datasets the accuracy metrics need to be assessed include F1-score along with Area Under the Curve (AUC).

The assessment of credit risk enhanced when Soni et al. [10] adopted contemporary ML and DL solutions instead of traditional statistical models based on Logistic Regression and Decision Trees. The study showed that Random Forests and Gradient Boosting with Dynamic Ensemble Selection present exceptional capability as ensemble methods to resolve data imbalance issues. The study assessed the merge between Information Gain and Principal Component Analysis (PCA) for preprocessing techniques that enhance operational outputs through better modeling outcomes. The paper illustrated SMOTE and additional class-balancing procedures as vital steps for developing reliable default prediction results. The authors have identified three types of prediction threats which include statistical threats and technological hazards together with market conduct risks that create conditions where instability and bias overlap with scalability and latency and ethical and regulatory compliance. Experimental studies demonstrate that combination

models utilizing XGBoost together with additional models produce top risk-adjusted results and superior forecasting performance which satisfies regulatory demands.

A new credit default prediction system presented by Lee et al., [4] utilized Graph Convolutional Networks (GCNs) to study borrower network connections for improving existing model limitations. Borrowers develop three categories of virtual distances using loan information combined with credit records as well as employment data and salary data. The authors created linked data networks from these simulated distances by making borrowers represent nodes and edges represent the connections. The system divided borrower evaluation between autonomous GCN models that use different types of data to discover both linear and complex relational patterns for understanding total behavioral patterns. Zedda [14] conducted an evaluation of logistic regression and XGBoost algorithm performance for credit scoring Italian SMEs. XGBoost demonstrated higher performance than standard tools and the study found logistic regression as a leading choice based on its straightforward interpretability features. The study integrated banking variables and financial ratios (28 + 55) to evaluate model success by utilizing an efficiency index which determines effective denial of worthwhile loans among good borrowers and unlawful granting loans to defaulters. Logistic regression produced trustworthy initial predictions regarding the target variable and XGBoost applied its performance-enhancing capabilities to certain circumstances thus expanding current predictive methods.

Ma et al. [5] investigated how XGBoost and LightGBM machine learning methods determine P2P default predictions within lending club transactions. The authors evaluated these algorithms using lending club transaction data by testing two high-dimensional cleaning methods referred to as multi-observation cleaning and multi-dimensional cleaning. The authors demonstrated that correct data preprocessing methods play a critical role in achieving well-performing predictive models for datasets with high dimensions of credit risk data. Yang et al. [13] deployed HDNN technology to address classification issues in corporate credit risk prediction tasks dealing with high-dimensional data databases. L1 and L2 regularization techniques were used by the researchers because they allowed for better feature selection while minimizing overfitting problems to enhance generalization effectiveness. The HDNN model achieved superior prediction results than standard Logistic Regression and Support Vector Machines implementations in experimental testing. The predictive capacity of the model increased through early detection of risk situations after incorporating external information from financial and supply chain sources and network data. According to Xu and Zhang [12] the research on Chinese technology-based enterprise financing risks sought to investigate the effects of information asymmetry restrictions

and financing decision doubt. The authors presented an evaluation system for monitoring credit risk by integrating profitability and debt-paying capability and operational efficiency together with growth potential and innovation capacities. The researchers implemented support vector machine (SVM) emphasizing fuzzy integral theory enhancement when integrating different financial performance indicators.

Korangi et al. [3] introduced a transformer-based deep learning algorithm which estimated default risks in mid-cap corporate sectors through processing of special dangers that mid-cap companies encounter during non-investment ratings. Traditional models convert timeseries information to static cross-sectional features but the authors created models through sequence data that recognizes temporal patterns spanning three months to three years. A combined method allows researchers to examine default risk understanding by combining data from financial reporting statements with business assessment surveys and market reaction data. The researchers developed static cross-sectional features through time-series aggregation before using them in logistic regression, XGBoost and Long Short-Term Memory (LSTM) network analysis. The transformer outperforms other forecasting methods across all prediction period durations because of its competence to effectively analyze complex temporal structures. Businesses that combine deep learning systems with big data platforms achieve improved rates of corporate credit risk prediction [11]. The prediction model incorporates financial metrics with non-financial elements through a data sample analysis of 1000 listed companies from 2012 to 2019. The research carried out by Xianyu and Hai [11] utilized Random Forest along with Logistic Regression algorithms together with deep learning method approaches. The authors verified RNN surpassed conventional predictive methods because it led to exceptional prediction results. The authors Xianyu and Hai [11] stressed that efficient capture of temporal dependencies within corporate data is crucial because RNN models demonstrate exceptional ability in sequential pattern-based credit risk assessments.

Moscatelli et al. [7] studied machine learning (ML) models' predictive functions for corporate default events using results from Logistic Regression (LOG) and Linear Discriminant Analysis (LDA). Analysis of default prediction patterns using Random Forest and gradient-boosted trees algorithms on Italian non-financial firm data from 2011 to 2017 was performed by the authors to determine the extraction of complex non-linear patterns. ML methods surpass traditional methods for public data analysis since they can extract valuable information from challenging multivariate data structures effectively. Through analyzing company earnings calls Siddiqui et al., [9] established differentiating points between firms which filed for bankruptcy versus those that did not become bankrupt. The research evaluated emotional expressions from CEO earnings calls because these

contain financial information about company health. A deep learning model processed information that originated from LSTM network features. CT-XGBoost represents a hybrid model which applies ensemble capabilities at the algorithm level to remedy class imbalance during energy sector credit default predictions according to Wang et al., [8]. The authors enhance XGBoost through their modifications that combine cost-sensitive methods with threshold adjustments to boost accuracy in minority default case identification. CT-XGBoost outperforms conventional XGBoost models as well as standard models like Logistic Regression, Support Vector Machines (SVM), Neural Networks and Random Forests when processing commercial bank data in western China. The research by Manurung et al. [6] estimated business bankruptcy risk during 2016–2018 especially for coal mining companies operating in Indonesia. The research project implemented Support Vector Regression for its modeling operations. Jemai and Zarrad [2] studied an engineering-based approach to feature selection through their analysis of UFS (Univariate Feature Selection), RFE (Recurisive Feature Elimination), FIDT (Flexible Identical Decision Tree) along with IV (Information Value).

The thorough literature review shows that machine learning and deep learning models were used in determining the creditworthiness of the companies over the period. The ensemble models and neural network models are found to be the most widely used and powerful algorithms when compared with the traditional machine learning algorithms and this study focuses on the comparison and analysis of the performance of different ensemble models and LSTM models after applying different feature engineering techniques.

Methodology of the Study

The dataset for this study was extracted from the CIME Prowess database, which contained financial information about a wide range of Indian companies. For data extraction, necessary queries were executed through the interface of the Prowess database (Prowessiq). Credit ratings of the company provided by ACUTE and BRICKWORK helped as the primary benchmarks for the study. To minimize bias in the dataset, a balanced selection of companies was ensured, which included an equal number of well-rated and poorly-rated companies. This resulted in a dataset comprising 198 defaulting companies and 212 high-quality non-defaulting companies from 2016 to 2024. The main objective of the study was to analyze various attributes that help in distinguishing a default company from non-defaulters by applying various machine learning algorithms and feature engineering techniques. Defaulting firms were labeled with a value of 1, while non-defaulting firms were labeled as zero. In addition to the data extraction, new features are generated

from the extracted features which were generally considered in determining the creditworthiness of a company. Some of the derived features were Market Capitalization/Enterprise value, Enterprise value/Revenue, Profit Margin, Operating cash flow ratio, Debt-to-equity ratio, Liquidity ratio, Return on equity ratio, and Leverage ratio. To avoid multicollinearity, after the creation of derived features, the base features that were used in the creation of derived features were removed from the dataset. As a result, the analysis was conducted using 27 distinct variables, as outlined in Table 1. The final dataset comprised 374 records and 28 columns of data.

The conceptual framework for the study is illustrated in Fig. 1. Dimensionality reduction techniques like Principal Component Analysis (PCA) and feature selection with a Random Forest Algorithm were used to extract the important features. Machine learning algorithms like XGBoost, CATBoost, etc. were employed to predict defaulters. To evaluate and determine the most effective model, various performance metrics such as accuracy, precision, recall, etc. were used.

Feature Engineering Techniques

In the first stage of the study, dimensionality reduction techniques such as PCA and feature selection based on variable importance from the Random Forest algorithm were used and important features were extracted.

Principal Component Analysis (PCA)

PCA is one of the most widely used dimensionality reduction techniques used in the reduction of the high dimensional dataset to low dimensional one while preserving as much variance (information) as possible. For the parameter $n_components=0.95$ which denotes that 95% of variance or information in data is retained and the features that don't satisfy the condition are considered insignificant features. In this study, a total of 21 components were created which is shown in Fig. 2.

Random Forest Algorithm for Feature Selection

The random forest algorithm is one of the most popular and robust ensemble algorithms used for both regression and classification problems. For this study, a random forest model was created through which most important features were selected. The model is created with the number of estimators initialized to 100 which denotes the number of decision trees that are required for the creation of the model. Here, the important features were selected based on the threshold value of 0.01 i.e. to keep the important features

with a value greater than 0.01. Important scores smaller than 0.01 are generally considered as noise or insignificant features and elimination of these features would help in improving the accuracy of the model. After the model creation and application of threshold value, out of 27, 19 features were considered important by the model. They are consolidated EPS, consolidated cash EPS, Yield, P/E, Turnover, BV per share, short-term payables and acceptance, Alpha, Net cash flow from investing activities, Profit margin, return on equity, Cash and cash equivalents at the end of each year, Market capitalization/Enterprise value, Debt-to-equity ratio, P/B, Non-current liabilities, Total Expense, return on equity, short term borrowings and Net cash flow from financing activities. Fig. 3 shows the feature importance graph of the random forest algorithm. Based on the outcomes, the features were selected and considered for building machine learning models.

Machine Learning Models

Machine learning models were built on the two versions of datasets, one comprising of components extracted from PCA and the second with features engineered using a random forest algorithm. Four different algorithms such as XGBoost, CATBoost, LightGBM, and LSTM were applied to the datasets. The performance of each model was compared using metrics such as accuracy, precision, recall, F1 score, and area under the curve (AUC) and analyzed to determine the best model for predicting companies that are likely to have credit risk.

XGBoost

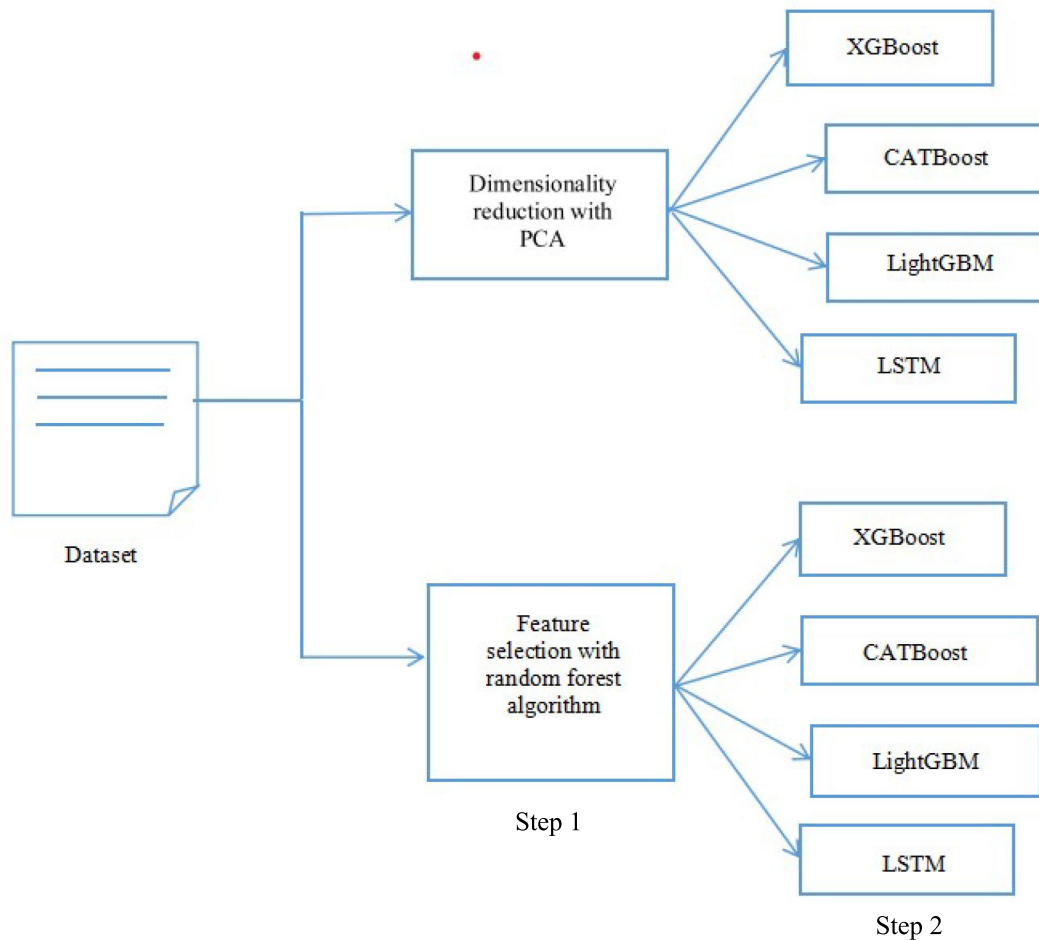
It is one of the most robust ensemble machine-learning algorithms available. XGBoost model was first applied to the components extracted using PCA. For the number of components necessary for the model creation, a percentage value (float value) is used which explains the variance (information) of the data. For the study, the standard value of 0.95 is used. Before, the application of PCA, data was scaled to get effective results from PCA. The application of PCA, resulted in the generation of 19 components which contributes to the overall accuracy (information) of the model. For the parameter settings of the model, for objective of the model, the value "binary:logistic" was specified since the nature of the study was binary classification (default or non-default) and for eval_metric, the value "logloss" was used which is also known as binary cross entropy which is most commonly used for binary classification problems. For the parameter max_depth which denotes the depth of each individual decision tree used in XGBoost, 4 was used. For the parameter eta which denotes the learning rate of the

Table 1 Independent variables with description

Predictors	Description
1. Total Expenses	The aggregate costs incurred by a business or individual over a specific period, including operating expenses, interest payments, and taxes
2. Net cash flow from investing activities	The cash gained or spent on investment-related activities, such as the purchase or sale of assets like property, equipment, or investments
3. Net cash flow from financing activities	The cash produced or used by financial activities like issuing stock, repurchasing shares, or repaying loans
4. Cash and cash equivalents as at the end of the year	The total cash and highly liquid assets held by a company at the end of the fiscal year
5. Non-current liabilities	Financial obligations that exceed one year's due date include long-term loans as well as deferred tax liabilities and bonds payable
6. Short-term borrowings	Debt obligations are scheduled for repayment within one year, typically including short-term loans or lines of credit
7. Short-term trade payables and acceptances	The outstanding amounts owed to suppliers or vendors for goods and services must be paid in a short time frame
8. Beta	A statistical measure of a stock's volatility indicates how much its price fluctuates relative to a benchmark index
9. Alpha	A measure of an investment's excess return compared to a market index, representing performance beyond what beta predicts
10. Total returns	The overall gain or loss from an investment, including both capital appreciation (or depreciation) and income from dividends or interest
11. Face value	The nominal or base value of a stock or security as determined by the issuer, is often used for accounting or to calculate dividends or interest
12. Consolidated EPS	The earnings per share accounts for the financial performance of a company and all its subsidiaries or affiliates
13. Consolidated cash EPS	A measure of earnings per share based on cash flow, indicating a company's ability to produce cash
14. P/E (Price-to-earnings ratio)	A metric that measures stock value to earnings per share data to understand market predictions for future financial performance
15. P/B (Price-to-book value ratio)	Stock market evaluation of a company's assets occurs through the equivalence of stock market price to book value per share
16. Book Value per share	The allocation of total equity divided by functioning shares reveals the theoretical value shareholders receive during company dissolution
17. Return	The earnings on an investment, are usually expressed as a percentage, including income from dividends, interest, or other sources relative to cost
18. Turnover	The speed at which a company converts inventory, receivables, or other assets into cash or sales revenue within a specific period
19. Market Capitalization/Enterprise value	A ratio that compares a company's market capitalization to its enterprise value, giving insights into its financial structure
20. Enterprise value/profit before income tax depreciation and amortization	A ratio comparing a company's enterprise value to its earnings before depreciation, interest, taxes, and amortization, measuring valuation relative to operations
21. Profit margin	A profitability ratio that shows the percentage of revenue left as a profit after covering all expenses
22. Operating cash flow ratio	A financial indicator that demonstrates a company's ability to pay current obligations using operational cash obtained from primary business operations
23. Debt-to-equity ratio	A financial metric that shows companies' debt level in comparison to shareholder equity while revealing their debt versus equity funding mix. A financial ratio examines the balance between debt funding and equity funding used for operating expenses
24. Liquidity ratio	A financial measurement technique companies utilize to evaluate their capacity to handle current liabilities by mobilizing liquid assets primarily comprising cash alongside other readily convertible assets
25. Return on equity	This profitability ratio reveals how well shareholder equity produces business profit. It measures the quantity of profit earned from each dollar invested by shareholders in their equity
26. Leverage ratio	This financial measurement reveals debt quantities as compared to the combined assets and equity of an organization. Financial leverage analysis assists in both debt proportion assessment and evaluation of payment capabilities

Table 1 (continued)

Predictors	Description
27 Enterprise value/Revenue	Enterprise value compared to total revenue constitutes this valuation metric. The metric determines investor willingness to pay per unit of revenue created

**Fig. 1** Theoretical framework of the study

```

[55]: print("Explained variance ratio:", pca.explained_variance_ratio_)
      print("Number of components:", pca.n_components_)

Explained variance ratio: [0.13059767 0.10625515 0.0787188  0.06651426 0.06095786 0.05158166
 0.04822249 0.04525658 0.04233601 0.03851301 0.03565306 0.03304371
 0.03175724 0.02919955 0.02801252 0.02693312 0.02433211 0.02197953
 0.02132157 0.01809334 0.01700028]
Number of components: 21

```

Fig. 2 Number of components generated by PCA and explained variance value for each component

model, 0.1 was used. For the number of rounds or iterations which is generally required for any boosting algorithms, an

optimal value of 100 was used. Based on the given configuration, the model was created and the performance metrics

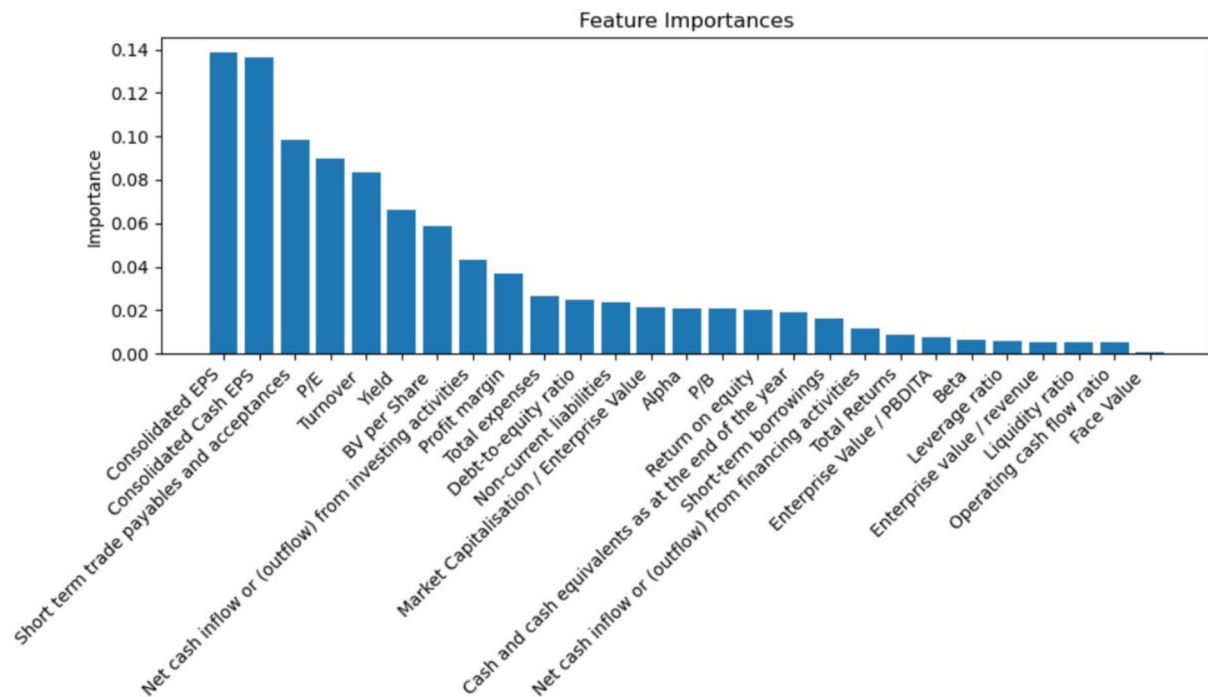


Fig. 3 Random Forest Variable Importance Plot

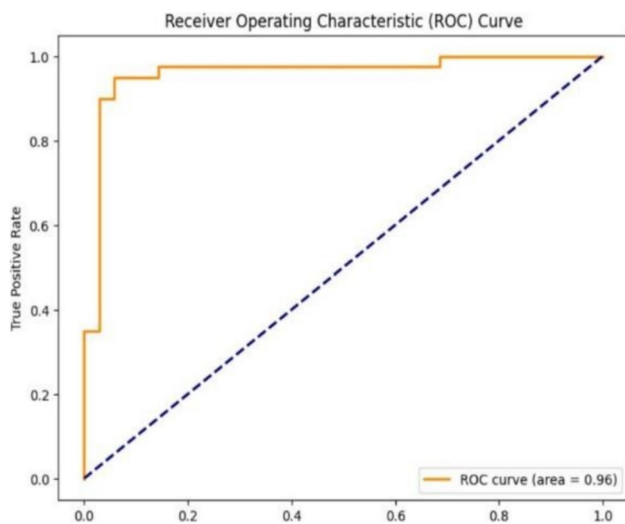


Fig. 4 ROC of XGBoost with PCA

were measured. The accuracy of the model is 92.10, precision value is 92.32, recall is 92.32, F1 score is 92.32 and the AUC score is 0.96. The AUC curve of the model built with PCA shown in Fig. 4.

The second model was built with the variables that were selected from the random forest with the same parameter configuration. The accuracy of the model is 98.68, precision is 100, recall is 97.45, F1-score is 98.8 and the AUC score is 1.00. The feature importance curve was created in order

to identify top 10 important features that would contribute to the accuracy of the model. The top 10 important features identified by the model were P/E, Short-term borrowings, consolidated cash EPS, Short term trade payables and acceptances, Alpha, Yield, Turnover, Market Capitalisation/Enterprise value, Net cash inflow or (outflow) from investing activities and profit margin. The ROC curve and the feature importance graph of the second model are shown in Fig. 5.

CATBoost

Similar to XGBoost, CATBoost is also one of the most powerful gradient boosting algorithms. The first model was created with PCA for dimensionality reduction. For retaining of variables or features that contribute to 95% of variance or information, for the parameter `n_features=0.95` was applied. This helps in the elimination of insignificant variables that won't contribute much in the model prediction. After the application of PCA, twenty-three components were created which contributes to 95% of the total variance of the dataset. The model was then created using the above components. The parameters such as the number of trees created was set 100, the depth set to 6 which indicates the depth of each tree, the learning rate set to 0.5 which denotes the step size for a stable training of the model, the `custom_metric` set to 'AUC' which is to measure the performance of the model and `verbose` set to 200 which refers to the verbosity of the model. The accuracy of the model is 93.33, precision is 94.74, recall

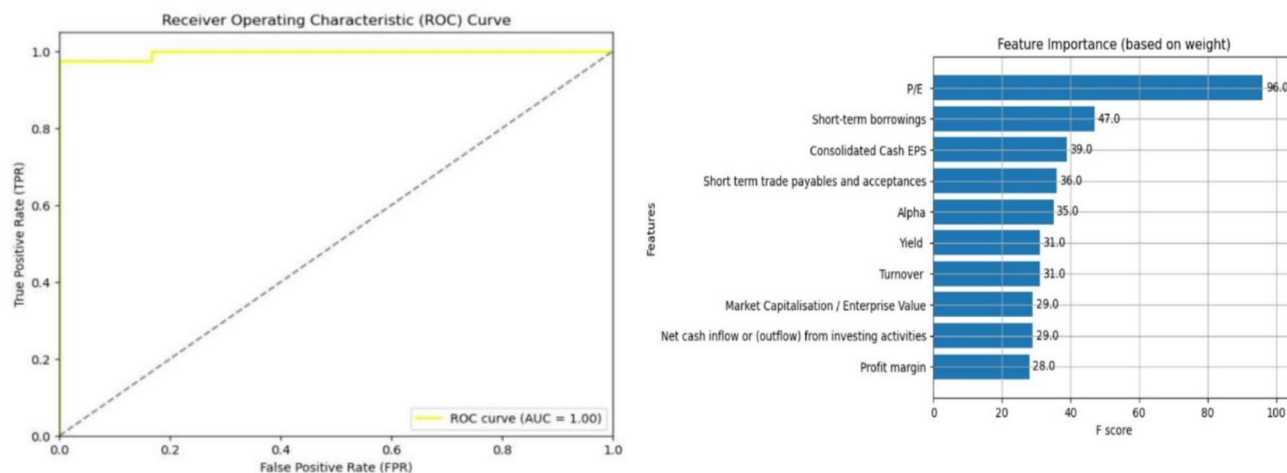


Fig. 5 ROC of XGBoost with feature selection from random forest and Feature Importance Plot

is 92.32, F1 score is 93.52 and the AUC curve is 0.97. The AUC curve of the model built with PCA shown in Fig. 6.

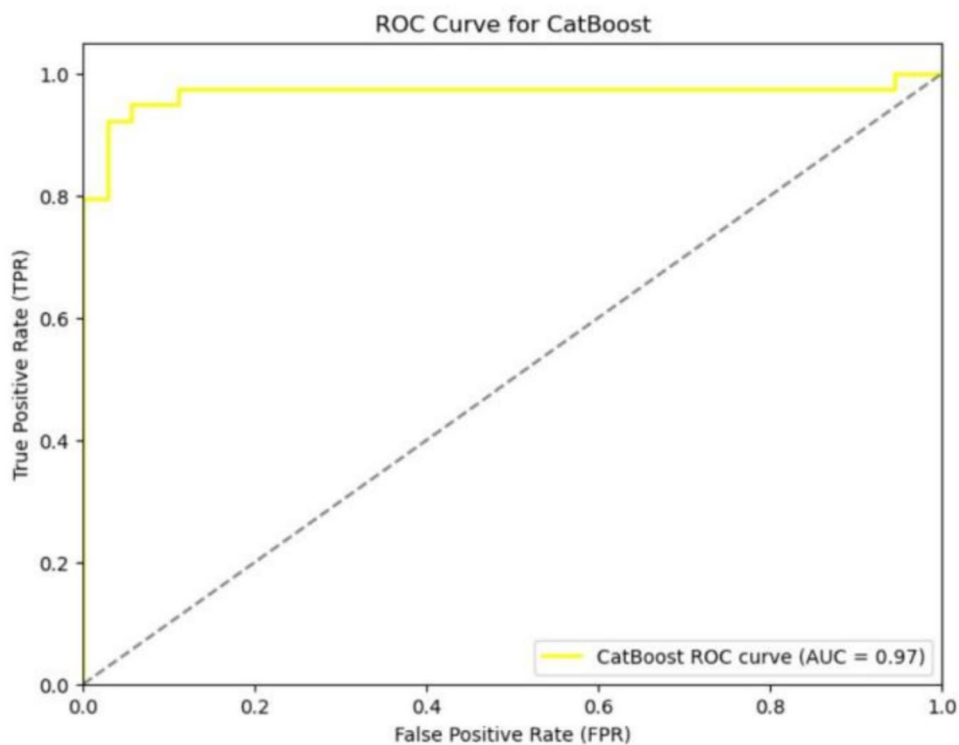
The second model was built with the important features identified from the random forest algorithm. The accuracy of the model is 98.67, precision is 100, recall is 97.44, F1 score is 98.7 and the AUC score is 0.99. The top 10 important features that were identified by the model are P/E, consolidated EPS, consolidated cash EPS, Yield, short term trade payables and acceptance, Turnover, Short-term borrowings, BV per share, Market capitalization/Enterprise value and net

cash inflow or (outflow) from investing activities. The ROC curve and the feature importance graph of the second model are shown in Fig. 7.

LightGBM

LightGBM is also one of the powerful gradient boosting algorithms that are similar to both XGBoost and LightGBM. The first model was created using PCA and twenty-three components were obtained from its application. The

Fig. 6 ROC Curve of CatBoost with PCA



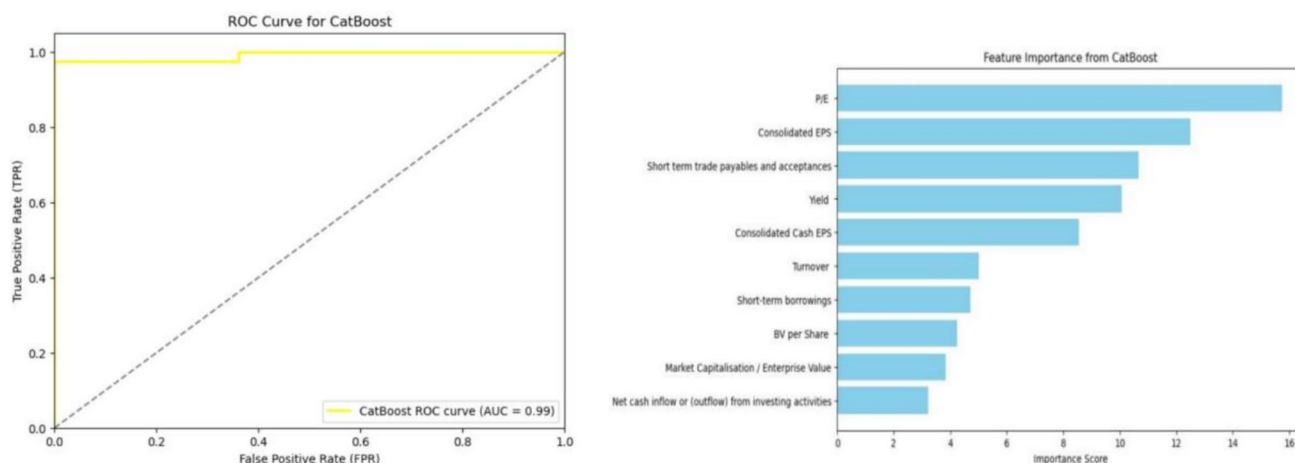


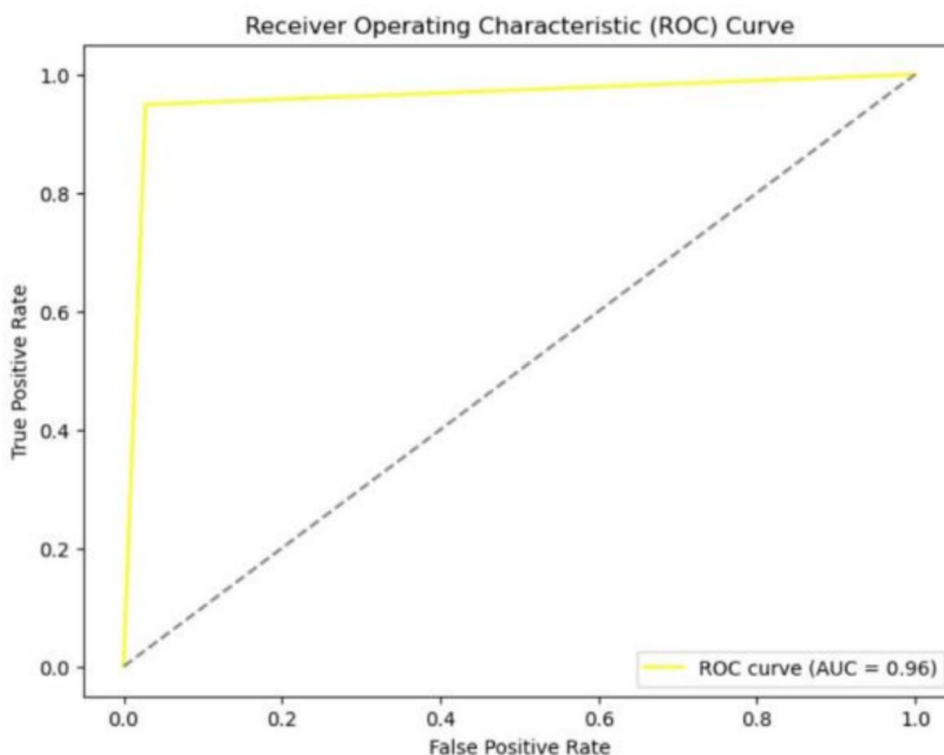
Fig. 7 ROC Curve of CATBoost with feature selection from random forest and Feature Importance Plot

parameters initialized for the creation of model were, objective which was initialized to binary, metric which was initialized to binary error, boosting type was initialized to gbdt which denotes the boosting algorithm used, num_leaves which denotes the maximum number of leaves for each decision tree which was initialized to 31, learning_rate which was initialized to 0.05 which denotes the step size and feature_fraction which denotes the fraction of parameters to be considered for each tree. The accuracy of the model is 96.00, precision is 97.37, recall is 94.87, F1 score is 96.10 and the

value of AUC curve is 0.965. The AUC curve of the model built with PCA shown in Fig. 8.

The second model was built with the important features identified from the random forest algorithm. The accuracy of the model was 94.67, precision was 100.00, recall was 89.74, F1-score was 94.59 and the ROC value is 0.96. The top 10 important features that were identified by the model were P/E, alpha, short term trade payables and acceptance, yield, consolidated cash EPS, consolidated EPS, net cash inflow or (outflow) from investing activities, Market capitalisation/

Fig. 8 ROC Curve of Light-GBM with PCA



Enterprise value, Turnover and cash cash equivalents at the end of the year. The ROC curve and the feature importance graph of the second model are shown in Fig. 9.

LSTM

Long Short-Term Memory (LSTM) is a type of Recurrent Neural Network (RNN) which is capable of handling sequential data more effectively when comparing with traditional RNNs. Unlike the ensemble models which works in identifying non-linear relationships in static datasets, LSTM is more suitable for financial applications where long-term dependencies and historical trends significantly influence the outcome such as credit risk. Its architecture includes a memory cell along with three gates such as input, forget, and output gates that control the flow of information, enabling the model to capture and maintain long-term dependencies. The model taken for consideration is of three layered with regularization applied at each layer and a final output layer with the sigmoid activation function applied which is suitable for binary classification. The model is then compiled using adam optimizer and with loss as binary cross entropy. The first model was built with the application of PCA which resulted in the creation of 23 components. Then the model was trained with epoch value set to 90, batch size as 64, validation split as 0.2 and verbose as 1. The accuracy of the model is 93, precision is 93, recall is 95, F1-score is 94 and the ROC value is 0.965. The AUC curve of the model built with PCA shown in Fig. 10.

For the second model created from the features selected through the Random Forest algorithm, the accuracy of the model is 72.5, the precision is 72, the recall is 78, the F1-score is 75 and the ROC score is 0.82. The top 10 important features identified by the model are total expenses,

short-term trade payables and acceptance, Net cash flow from financing activities, non-current liabilities, cash and cash equivalents at the end of the year, consolidated EPS, BV per share, Net cash flow from investing activities, consolidated cash EPS and P/E. The ROC is shown in Fig. 11 and the feature importance curve is shown in Fig. 12

Results and Discussion

The results indicate that ensemble methods such as XGBoost, CATBoost, and LightGBM with the features selected by random forest perform better when compared with the models created with PCA. In the case of LSTM, the model created with the application of PCA performs better when compared with the model created through feature selection with a random forest algorithm. The summary of the performance of all the models is given in Table 2

The top 10 predictor variables of all the models created through features selected from the Random Forest Algorithm are tabulated (Table 3) which significantly influences in determining the prediction of a company's creditworthiness.

The increase of non-performing assets (NPAs) throughout the financial sector underlines the pressing requirement for robust credit risk management systems. For stakeholders including lenders along with management teams and credit analysts, regulators and investors, implementation of strong credit risk management helps in making better financial decisions. Future default prediction along with critical risk determinant identification emerged as the main advantage of predictive credit risk models. With these models, lenders can achieve proper credit selection for funding needs and generate optimal loan portfolio distributions. The financial decision-making process for

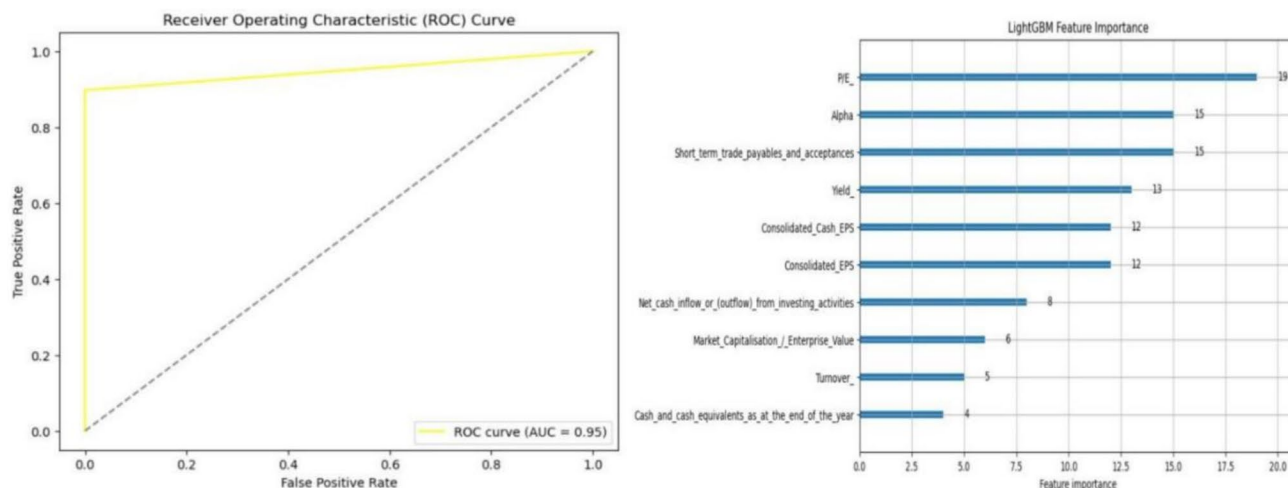


Fig. 9 ROC Curve of Light with feature selection from random forest and Feature Importance Plot

Fig. 10 ROC Curve of LSTM with PCA

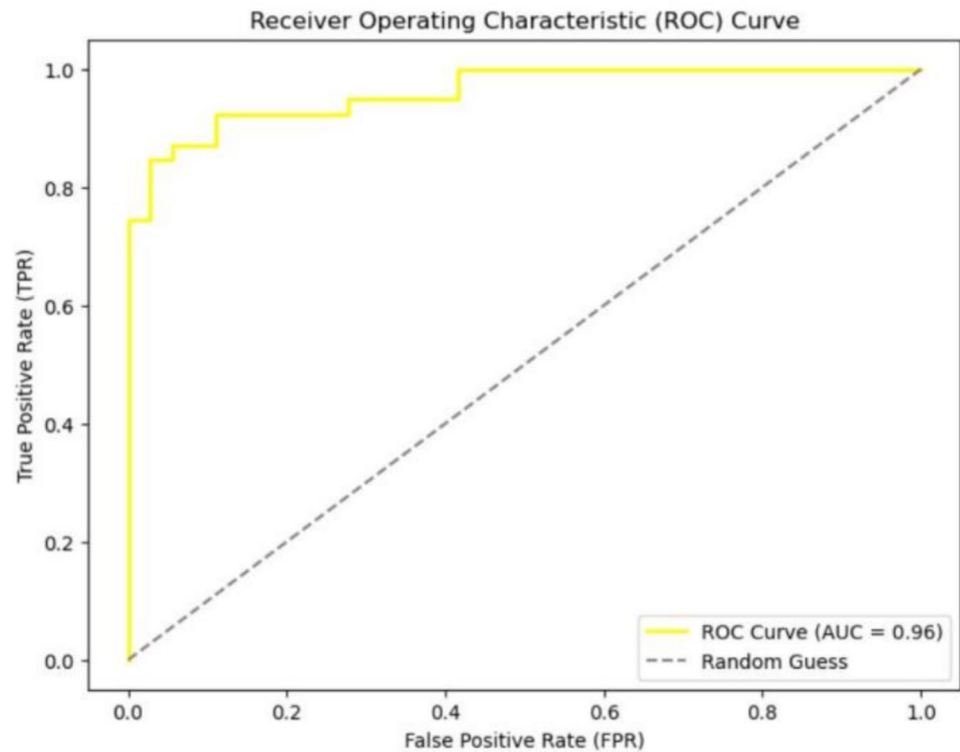
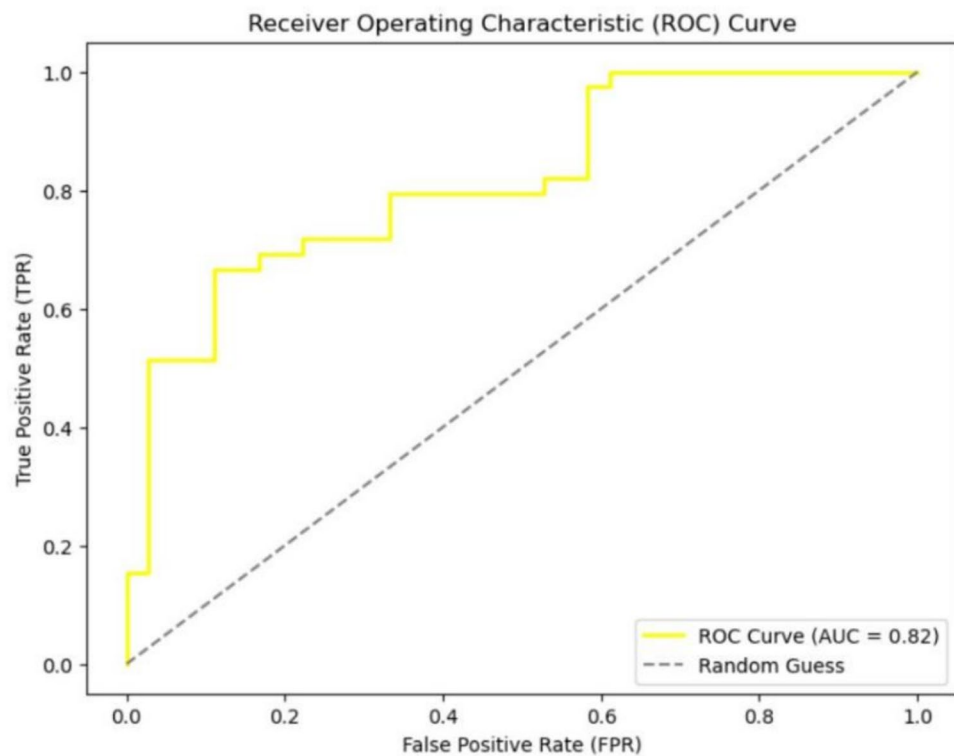


Fig. 11 ROC Curve of LSTM with feature selection from Random forest



corporate management, investors and regulatory institutions benefits from these models. Financial evaluation functions more efficiently because this produces both

better performance and fewer collection problems and strengthens organizations' credit risk processes. The financial distress prediction models help companies assess their

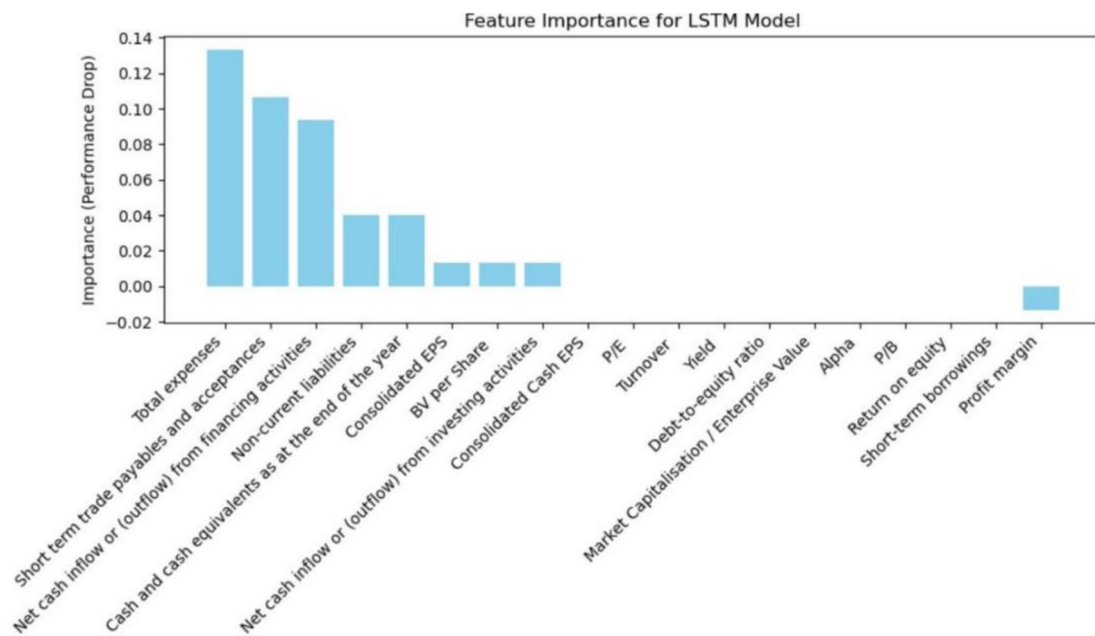


Fig. 12 Feature importance plot for LSTM with feature selection from random forest

Table 2 Summary of Results

Model	Dimensionality reduction techniques	Accuracy	Precision	Recall	F1 Score	AUC
XGBoost	PCA	92.1	92.32	92.32	92.32	0.96
	Feature selection with random forest	98.68	100	97.45	98.8	1
CATBoost	PCA	93.33	94.74	92.32	93.52	0.97
	Feature selection with random forest	98.67	100	97.44	98.7	0.99
LightGBM	PCA	96.00	97.37	94.87	96.10	0.965
	Feature selection with random forest	94.67	100	89.74	94.59	0.96
LSTM	PCA	93	93	95	94	0.965
	Feature selection with random forest	72.5	72	78	75	0.82

Table 3 Top 10 predictor variables in each model

Features selected through Random forest	XGBoost	CATBoost	LightGBM	LSTM
Consolidated EPS		*	*	*
Consolidated cash EPS	*	*	*	*
Yield	*	*	*	*
P/E	*	*	*	
Turnover	*	*	*	
BV per share		*		*
short term payables and acceptance	*	*	*	*
Alpha	*		*	
Net cash flow from investing activities	*	*	*	*
Profit margin	*			
Total expense				*
Cash and cash equivalents at the end of each year			*	*
Market Capitalization/Enterprise value	*	*	*	
Non-current liabilities				*
Short term borrowings	*	*		
Net cash flow from financing activities				*

financial status even before crises begin while recognizing preliminary economic warning signs to apply proactive solutions.

Practical Implications

The machine learning model enhance the credit risk assessment processes in banks and other NBFCs (Non-Banking Financial Services) by accurately predicting whether a company becomes default or non-default, thereby reducing the non-performing assets of the financial institutes. In addition, investment and portfolio managers could use these models to assess the financial health of the companies before making investment decisions. Government and other credit rating bureaus could use these models to identify financial risk across industries and could take necessary actions to ensure financial stability.

Conclusion

This study utilized two feature engineering techniques and four machine learning models for the prediction of the creditworthiness of a company. The research utilized feature engineering techniques like Principal Component Analysis which reduces the dimensions or features of the dataset to avoid complexity and further used feature selection based on the Random Forest algorithm to identify the best or potential features. Furthermore, the study utilized four machine learning techniques namely XGBoost, LightGBM, CATBoost, and LSTM. Two models were created using each machine-learning technique. One was trained with the components generated from PCA and the other with the features selected by the random forest algorithm. The predictive power of the two models created using different engineering techniques for each algorithm was compared with evaluation metrics such as AUC, ROC curves, precision, recall, and F1 score. The results showed that the accuracy is highest for ensemble models like XGBoost, LightGBM and CATBoost which were created through features that were selected from the Random Forest compared to the same models created through PCA. In the case of LSTM, the model created with PCA has shown higher accuracy compared with the other model created from feature selection. The important variables identified by most of the models were Consolidated EPS, Consolidated Cash EPS, P/E, yield, short-term payables, turnover, net cash flow from investing activities, market capitalization/Enterprise value and acceptance, and short-term borrowings.

Limitations and Scope for Future Research

The study could be expanded further by including other potential variables or features that would significantly influence the credit rating of the companies. Many other feature engineering techniques can also be explored and applied for further research. Multi-class classification algorithms could also be explored by including medium or mid-level companies since the study focuses only on binary classification and considers only high-security and default companies.

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Data Availability The data supporting this study's findings are available from the corresponding author upon request.

Declarations

Conflict of Interest The authors declare that they have no known competing interests or personal relationships that could have appeared to influence the work reported in this article

Research Involving Humans and/or Animals This study did not involve human participants or animals.

Informed Consent Informed consent was not required as this study did not involve human participants.

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